

MODEL QUESTION PAPER

TED (10)-4084
(REVISION-2010)

Register Number.....
Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

(Common to EI & IT)

PROCESS CONTROL

(Maximum marks: 100)

[Time: 3 Hours]

PART – A

I answer all questions. Each question carries two marks.)

1. Define control lag in process control.
2. Name four process characteristics.
3. List four elements used in implementing pneumatic controller.
4. Write the uses of voltage follower.
5. State cavitation of control valve?

(5 x2 = 10 Marks)

PART – B

II Answer any five questions. Each question carries six marks.

1. Describe self regulation with relevant example.
2. A controller outputs a 4 to 20 mA signal to control motor speed from 140 to 600 rpm with a linear dependence. Calculate
 - (a) current corresponding to 310rpm and
 - (b) the value of (a) expressed as the percentage of control output.
3. Implement and describe pneumatic PI controller.
4. Describe single speed floating control mode.
5. Compare PI and PD control modes.
6. Summarize the characteristics of an ideal op-amp.
7. Differentiate air to open and air to close control valves.

(5x6 = 30 Marks)

PART C

Answer one full question from each module. Each question carries 15 marks.

MODULE -I

- III (a) With the help of block diagram process control loop. (7)
(b) Describe a temperature control system with schematic diagram. (8)

OR

- IV (a) Write short notes on control system parameters. (8)
(b) Describe a Pressure process control system with block diagram. (7)

MODULE –II

- VI (a) Describe flapper-nozzle system with neat diagram. (6)
(b) Compare PI, PD and PID Controller modes. (6)
(c) Write advantages of two position control mode. (3)

OR

- VII (a) Describe pneumatic PID controller with neat sketch. (8)
(b) A proportional Pneumatic controller has $A_1=A_2= 5 \text{ sq.cm}$, $x_1=8\text{cm}$, $x_2 = 5\text{cm}$. Input and Output pressure ranges are 3 to 15 psi. Find the input pressures that will drive the output from 3 to 15 psi. Set point pressure is 8 psi and $p_0= 10 \text{ psi}$. Find the proportional Band. (7)

MODULE –III

- VIII (a) Draw and explain the instrumentation amplifiers using Operational Amplifier. (7)
(b) Explain the working of electronic PID controller. (8)

OR

- IX (a) Describe Inverting and Non-inverting amplifier with suitable diagram. (8)
(b) Construct a Voltage to Current Converter using operational amplifier and explain. (7)

MODULE –IV

- X (a) Explain working of pneumatic actuators with diagram. (7)
(b) Describe the functions of valve positioner and motion transmitter. (8)

OR

- XI (a) Describe principle operation of solenoid valve (6)
(b) Explain the direct acting and reverse acting of control valve (6)
(c) Write the principle of operation of pneumatic actuator. (3)
